

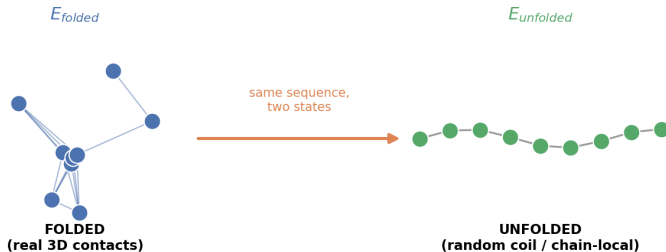
DeepPEF v5 — The Six Levers (A–F)

Physics-grounded, flag-gated, ablatable. Default = OFF = baseline.

Teaching deck

The core idea

The core idea: folding free energy as an energy difference



$$\Delta G = E_{unfolded} - E_{folded} \quad \Delta\Delta G = \Delta G_{mut} - \Delta G_{wt}$$

A

folded graph (real 3D contacts) vs an unfolded reference (chain-local). $\Delta G = E_{unfolded} - E_{folded}$;
 $\Delta\Delta G = \Delta G_{mut} - \Delta G_{wt}$.

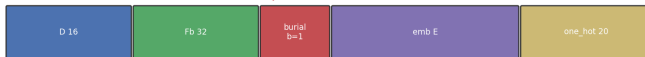
The dimension contract

Node-feature layout & where each lever changes it

BASILINE



+ Lever C (burial)



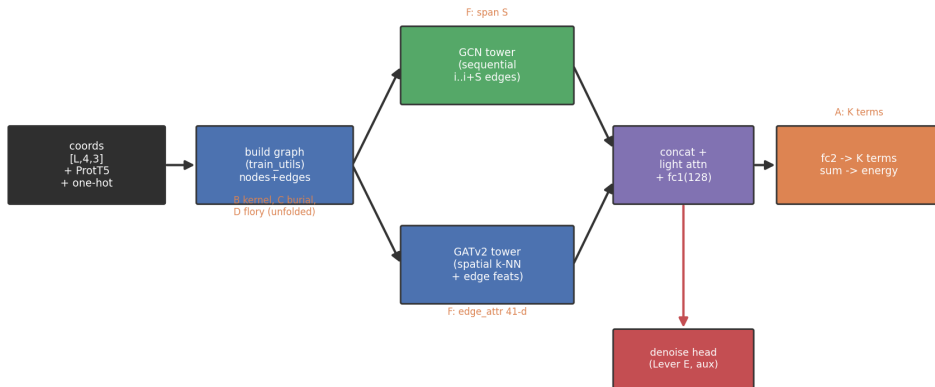
one_hot always stays LAST (model slices it from the right).
Burial is inserted BEFORE emb; every width derives from CFG so nothing drifts.

Every width derives from

ONE CFG. Builder and model read the same numbers, so a lever's default reproduces the baseline tensors bit-for-bit.

The two-tower architecture

Two-tower architecture: from graph to per-residue energy



The six levers at a glance

The six levers (default = OFF = baseline)

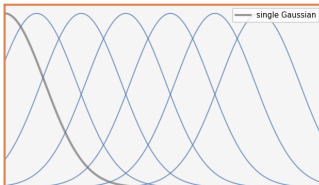
A Energy decomposition

fc2: 128 -> K per-residue terms



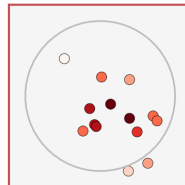
$K=1$ == today's single scalar

B RBF distance bank



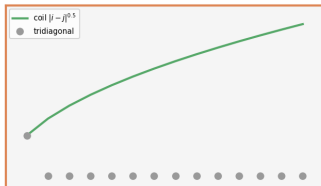
M RBFs summed -> width 16 (dim-preserving)

C Burial / solvation



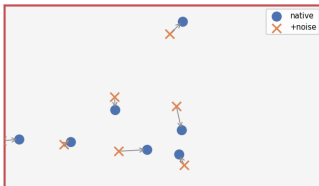
buried (dark) vs surface (light)

D Flory unfolded reference



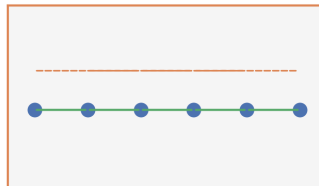
random-coil distances (value-only)

E Decoys + denoising head



head predicts injected noise (train only)

F Edge features + connectivity



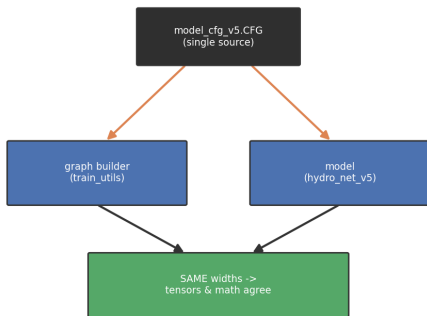
span 5 both dirs; edge_attr=[oh|oh|dist]=41

What each lever does

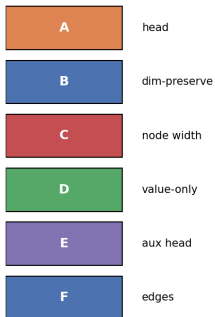
Lever	Flag	Effect (default reproduces baseline)
A	<code>--energy_terms K</code>	fc2 emits K per-residue energy terms; K=1 = scalar.
B	<code>--rbf_centers M</code>	M RBFs summed back to width 16 (dimension-preserving).
C	<code>--use_burial</code>	CB neighbor-density scalar inserted before emb.
D	<code>--flory_unfolded</code>	random-coil unfolded reference $d \sim i - j ^\nu$ (value-only).
E	<code>--denoise_weight w</code>	aux head predicts injected coord noise (train only).
F	<code>--gcn_span S, --use_edge_features</code>	span-S GCN edges + 41-d GAT edge features.

Why the levers compose

The dimension contract



Composability (why levers stack cleanly)



A/E orthogonal to widths | B preserves width | D value-only
C changes node width, F changes edges -> compose via CFG

- CPU smoke test runs baseline **and each lever** on synthetic data — no GPU, no real data.
- Baseline regression: default flags $\Rightarrow$ no denoise/edge params, `fc2=(1,128)`, `burial_dim=0`, `gcn_span=1`.
- Fail-fast checks name the offending lever on any bad config or width mismatch.
- GPU ablation ladder measures PCC one lever at a time.

`python DeepPEF_v5/tests/smoke_test.py` $\rightarrow$ all configs PASS.

Evaluation metrics

- **PCC** (primary): $r = \text{cov}(p, y) / (\sigma_p \sigma_y)$ — linear trend, shift/scale-invariant. Historical best ≈ 0.5259 .
- **Spearman** ρ : Pearson on ranks — monotonic ordering, outlier-robust.
- **RMSE** $= \sqrt{\text{mean}((p - y)^2)}$ — absolute error (diagnostic; energy is in relative units, not calibrated kcal/mol).

Correlation grades the *pattern* (what matters for ranking variants); RMSE grades *calibration*. We report all three.

$$\text{loss} = \text{primary} + \text{reg} + \text{energy_reg} + \text{rank} + \text{corr} (+ \text{denoise})$$

- **Huber** ($\delta=1$): MSE near zero, L1 far out — robust to noisy labels.
- **Margin ranking**: $\max(0, -s \cdot \Delta p + \text{margin})$ — optimizes pairwise order.
- **Energy reg** $0.001 \|E\|^2$ — anchors the extensive energy scale.
- **Pearson loss** $1 - r$; **CCC loss** $1 - \text{CCC}$ (also penalizes shift + scale).
- **Denoise MSE** (Lever E) — consistency under structural noise.

- **Tsuboyama et al. 2023** mega-scale stability dataset.
- **K50 cDNA-display proteolysis**: folded proteins resist protease; K50 = cleavage midpoint $\rightarrow$ folding free energy dG . Higher K50 = more stable.
- **Clamp** $dG \in [-1, 5]$ kcal/mol (`--dg_ml`) — the assay's dynamic range.
- **Single mutations + PNAS curation** (`--one_mut`) — quality $>$ quantity; unfiltered data *hurt* performance.
- **ThermoMPNN** proteins + homologs held out — prevents test leakage.

EBM pretraining & the physics bet

- **Native = energy minimum:** shape the landscape so the native fold is lowest.
- **InfoNCE / Boltzmann:** $L = E_{\text{native}}/\tau + \log \sum_k e^{-E_k/\tau}$ (negatives: decoy seq/structure, unfolded, cyclic perms).
- **DSM:** teach ∇E to point back to native under noise ($v \cdot \nabla E \approx v \cdot (-\text{noise}/\sigma^2)$) $\rightarrow$ Lever E at fine-tuning.

Physics bet: fold–unfold difference + extensive (summed) energy + Flory reference shrink the hypothesis space to physically plausible functions — for data efficiency, interpretability, and generalization.

Limitations & scope

- No inverse-folding pretraining in baseline (largest known lever; the gap to ThermoMPNN).
- Fixed WT structure per protein — no mutation-induced local relaxation.
- Extensive energy confounds length unless `length_norm` is on.
- Relative-unit energy — RMSE is a diagnostic, not calibrated kcal/mol.
- Glycine virtual-CB caveat; ESM-IF 1536-d alignment relies on the CFG contract.